Here's a comprehensive table listing **Kubernetes services** categorized into:

- **Built-in Free Services** (part of open-source Kubernetes)
- **Premier Paid Services** (offered by cloud providers or third-party vendors)

Each service includes its **category**, **name**, **free/paid**, **provider**, and a **suggested use case**.

Kubernetes Services Table: Built-in (Free) vs. Premier (Paid)

Category	Service	Free/Paid	Provider	Suggested Use Case
Core Services	Kube-apiserver	Free (Built-in)	Kubernetes	Handles all API requests to the Kubernetes cluster
	Kube-scheduler	Free (Built-in)	Kubernetes	Assigns pods to nodes based on resource availability
	Kube-controller-manager	Free (Built-in)	Kubernetes	Handles controllers like node, replication, etc.
	etcd	Free (Built-in)	Kubernetes	Stores all cluster data in key-value format
	kubelet	Free (Built-in)	Kubernetes	Runs on each node, manages container lifecycle
	kube-proxy	Free (Built-in)	Kubernetes	Network proxy, load balances across pods

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Cluster Management & Monitoring

Category	Service	Free/Paid	Provider	Suggested Use Case
Monitoring	Metrics Server	Free (Built-in)	Kubernetes	Collects resource usage data (CPU, memory) for autoscaling
Monitoring	Prometheus	Free	CNCF / Open Source	Custom monitoring and alerting for Kubernetes
Monitoring	Grafana	Free	Grafana Labs	Visualizing Kubernetes metrics
Logging	Fluentd	Free	CNCF / Open Source	Log shipping to Elasticsearch, Stackdriver, etc.

Logging	ELK Stack	Free	Elastic	Centralized logging for Kubernetes
Logging	Loki	Free	Grafana Labs	Lightweight logging stack for Kubernetes
Paid Monitoring	Datadog Kubernetes Monitoring	Paid	Datadog	Full-stack monitoring with Kubernetes integration
Paid Monitoring	New Relic Kubernetes Integration	Paid	New Relic	Real-time performance monitoring
Paid Monitoring	Dynatrace Kubernetes Module	Paid	Dynatrace	AI-powered observability for Kubernetes

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Managed Kubernetes Services (Cloud Providers)

Service Name	**Free/Paid**	**Cloud Provider**	**Suggested Use Case**
Amazon EKS (Elastic Kubernetes Service)	Paid	AWS	Fully managed Kubernetes for production workloads
Google Kubernetes Engine (GKE)	Free Tier + Paid	Google Cloud Platform	Scalable, managed Kubernetes with optional autopilot mode
Azure Kubernetes Service (AKS)	Free + Paid Nodes	Microsoft Azure	Managed Kubernetes with deep Azure integration
DigitalOcean Kubernetes (DOKS)	Paid	DigitalOcean	Simplified Kubernetes for small to mid-scale apps
IBM Cloud Kubernetes Service	Paid	IBM Cloud	Enterprise-grade Kubernetes with Watson & ML integrations
Oracle Container Engine for Kubernetes (OKE)	Free Control Plane + Paid Nodes	Oracle Cloud	Kubernetes for Oracle workloads

Alibaba Cloud Container Service	Paid	Alibaba Cloud	Kubernetes tailored for Asian markets

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Security & Policy Services

Category	Service	Free/Paid	Provider	Suggested Use Case
Network Policy	Calico	Free	Tigera	Secure pod communication with network policies
Identity	Kubernetes RBAC	Free (Built-in)	Kubernetes	Control user and app access via roles
Admission	OPA/Gatekeeper	Free	Open Policy Agent	Enforce policies like restricting image registries
Container Security	Falco	Free	Sysdig	Runtime threat detection for Kubernetes
Paid Security	Prisma Cloud (Twistlock)	Paid	Palo Alto Networks	Container and Kubernetes security
Paid Security	Aqua Security	Paid	Aqua Security	Image scanning, runtime protection

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CI/CD and GitOps Services

Service	Free/Paid	Provider	Suggested Use Case
Argo CD	Free	CNCF / Open Source	Declarative GitOps CD for Kubernetes
FluxCD	Free	CNCF / Open Source	GitOps-based continuous delivery
Jenkins X	Free	Jenkins Community	CI/CD pipelines with Kubernetes native features
Skaffold	Free	Google	Easy local development and deployment
Tilt	Free	Tilt Dev	Streamlined Kubernetes dev workflow

GitLab Auto DevOps	Free + Paid	GitLab	CI/CD pipelines for Kubernetes using GitLab
Harness CD	Paid	Harness	No-code CD pipelines for Kubernetes

Service Mesh & Ingress

Category	Service	Free/Paid	Provider	Suggested Use Case
Ingress	NGINX Ingress Controller	Free	Open Source	Route external traffic to Kubernetes services
Ingress	Traefik	Free	Traefik Labs	Dynamic ingress controller with monitoring
Service Mesh	Istio	Free	Istio Community	Advanced service mesh for security, traffic management, observability
Service Mesh	Linkerd	Free	CNCF	Lightweight, simple service mesh
Service Mesh	Consul	Free + Paid	HashiCorp	Multi-cloud service mesh and discovery
Paid Service Mesh	AWS App Mesh	Paid	AWS	Managed service mesh for EKS or ECS
Paid Ingress	GKE L7 Internal Load Balancer	Paid	Google Cloud	Production-ready L7 load balancer for Kubernetes apps

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Summary

Type	Examples
Built-in Kubernetes Services	kube-apiserver, kubelet, controller-manager, etcd, kube-proxy
Open Source (Free Add-ons)	Prometheus, Grafana, ArgoCD, Calico, Fluentd, NGINX Ingress
Managed Cloud Services	EKS (AWS), GKE (Google), AKS (Azure), DOKS (DigitalOcean)

Paid Enterprise Tools	Datadog, Prisma Cloud, Aqua Security, New Relic

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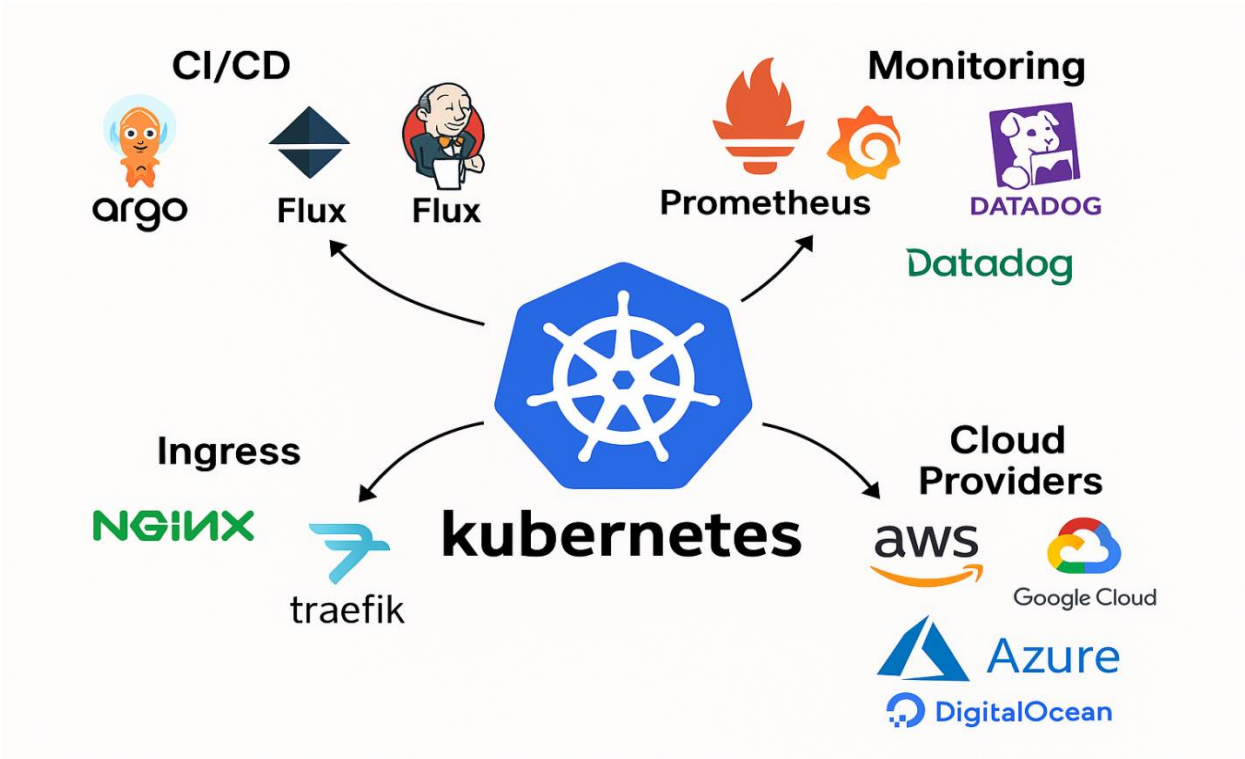
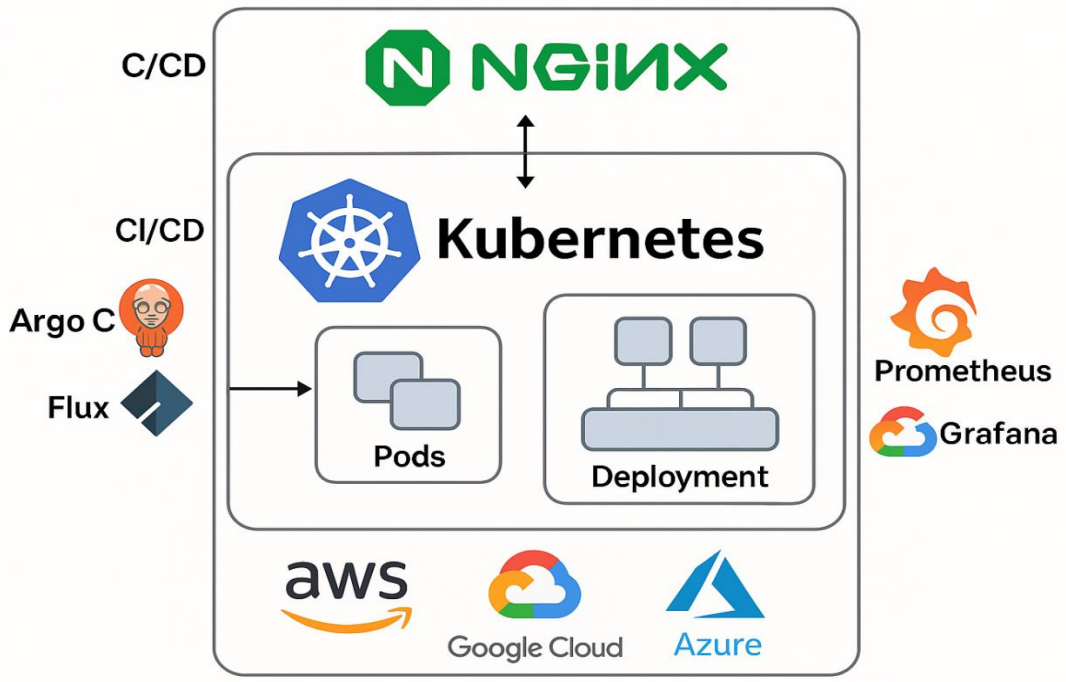
Would you like an **architecture diagram** showing how these services fit in a typical Kubernetes setup (CI/CD + Monitoring + Ingress + Cloud)?




C/CD
NGINX
CI/CD
Kubernetes
Argo C
Flux
Pods
Deployment
Prometheus
Grafana
aws
Google Cloud
Azure